EE4-29
EE9-CS3-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected Copy

OPTIMIZATION

Thursday, 30 April 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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OPTIMISATION

1. Consider the function

$$f(x) = (x_1 - 2)^4 + (x_1 - 2)^2 x_2^2 + (x_2 + 1)^2.$$

a) Compute the unique stationary point x_* of the function f. [2 marks]

b) Using second order sufficient conditions of optimality show that the stationary point determined in part a) is a local minimizer. Hence, show that f is radially unbounded and that the stationary point determined in part a) is the global minimizer of f. [4 marks]

c) Write the modified Newton's iteration for the minimization of the function f given by

$$x_{k+1} = x_k - \left[\nabla^2 f(x_*)\right]^{-1} \nabla f(x_k).$$

[4 marks]

d) Run five steps of the modified Newton's iteration in part c) from the starting point $(1.5, 0)$. [4 marks]

e) Run four steps of the modified Newton's iteration in part c) from the starting point $(1, 0)$. [2 marks]

f) Show that the research directions generated by the modified Newton's iteration in part c) are descent directions satisfying the condition of angle. Explain why the iteration is not globally convergent. [4 marks]

2. Consider the function

$$f(x) = \frac{1}{2}x_1^2 + \frac{m}{2}x_2^2,$$

with $m > 0$. The function has a global minimizer at $x_\star = 0$.

a) Show that the gradient algorithm with exact line search for the function f can be written as

$$x_{k+1} = x_k - \frac{x_{1,k}^2 + m^2 x_{2,k}^2}{x_{1,k}^2 + m^3 x_{2,k}^2} \begin{bmatrix} x_{1,k} \\ m\, x_{2,k} \end{bmatrix}$$

[6 marks]

b) Let $m = 9$ and $x_0 = [9, 1]^T$. Show that the sequence of points generated by the gradient algorithm is given by

$$x_k = \begin{bmatrix} 9 \\ (-1)^k \end{bmatrix} (0.8)^k.$$

(Hint: assume that for the given values of m and x_0 the quantity

$$\frac{x_1^2 + m^2 x_2^2}{x_1^2 + m^3 x_2^2}$$

remains constant for all iterations of the algorithm.)

[8 marks]

c) Compute the speed of convergence of the sequence generated by the algorithm and in particular show that

$$\frac{\|x_{k+1} - x_\star\|}{\|x_k - x_\star\|} = \text{constant}$$

for every k, where $\|v\| = \sqrt{v^T v}$. [6 marks]

3. Consider the optimization problem

$$\min_{x_1,x_2} 2x_1^2 + 9x_2,$$

$$x_1 + x_2 \geq 4.$$

a) State first order necessary conditions of optimality for such a constrained optimization problem. [4 marks]

b) Using the conditions derived in part a) compute candidate optimal solutions. [4 marks]

c) This constrained optimization problem can be transformed into an unconstrained optimization problem by defining the so-called barrier function

$$B_r(x) = 2x_1^2 + 9x_2 + r\frac{-1}{4 - x_1 - x_2},$$

with $r > 0$. Determine the stationary points of the function B_r. Show that there is only one stationary point in the admissible set, for all $r > 0$, and that this stationary point converges to the candidate optimal solution determined in part b). [6 marks]

d) By comparing the necessary conditions of optimality for the constrained optimization problem and the necessary conditions of optimality for the minimization of the function B_r, compute the optimal multiplier of the problem and show that this coincides with the value determined in part b). [6 marks]

4. Consider the optimization problem

$$\min_{x_1,x_2} 2(x_1^2+x_2^2-1)-x_1,$$
$$x_1^2+x_2^2-1=0.$$

This is a well-known problem which is used to illustrate the so-called Maratos effect: the slow convergence of algorithms exploiting the linearization of the constraints.

a) State first order necessary conditions of optimality for this constrained optimisation problem. [2 marks]

b) Using the conditions derived in part a) compute candidate optimal solutions. [4 marks]

c) Using second order sufficient conditions of optimality determine the solution x_* of the optimization problem and the corresponding optimal multiplier λ_*. [4 marks]

d) Consider a point $x_k=(x_{1,k},x_{2,k})=(\cos\theta_k,\sin\theta_k)$. This point is a feasible point. To determine an update for the point x_k one has to update the value of θ_k. This can be done using the following procedure.

i) Show that the linearization of the constraint around the point x_k is the constraint

$$2\cos\theta_k x_1+2\sin\theta_k x_2-2=0.$$

[2 marks]

ii) Consider the constrained minimization problem given by

$$\min_{x_1,x_2} 2(x_1^2+x_2^2-1)-x_1,$$
$$2\cos\theta_k x_1+2\sin\theta_k x_2-2=0.$$

Using first order necessary conditions of optimality determine a candidate optimal solution

$$x_*(\theta_k)=\begin{bmatrix} x_{1,*}(\theta_k) \\ x_{2,*}(\theta_k)\end{bmatrix}.$$

[4 marks]

iii) Consider the update law

$$\theta_{k+1}=\left(\frac{x_{2,*}}{x_{1,*}}\right),$$

with $x_{1,*}(\theta_k)$ and $x_{2,*}(\theta_k)$ as determined in part d.ii). Let $\theta_0=0.1$ and run five iterations of the algorithm. Argue that the sequence converges to $\theta=0$ with linear speed of convergence. [4 marks]